

SECONDARY SCIENCE EXAMINATION- 2019

(ANNUAL)

SCIENCE

TOTAL NO. OF QUESTIONS: 68

[TIME: 2 HOURS 30 minutes]

F. M. 80

Instructions for Candidates:

SECTION – A

Objective Type questions

Question no.-1 to 40 have four options out of which one is correct. YOU have to mark your selected answer on OMR sheet. [40 × 1]

1. The loss of oxygen from a substance during a chemical reaction is called?

- (A) Oxidation (B) Reduction
(C) Corrosion (D) None of these

Ans: Reduction

2. Food cans are coated with tin not with Zinc because

- (A) Zinc is costlier than Tin (B) Zinc has a higher melting point than Tin
(C) Zinc is more reactive than Tin (D) Zinc is less reactive than Tin

Ans: Zinc is more reactive than Tin

3. Which substance changes colour of red litmus to blue?

- (A) Acid (B) Base
(C) Salt (D) None of these

Ans: Base

4. Which Gas is responsible for Global Warming?

- (A) Carbon dioxide (B) Oxygen
(C) Nitrogen (D) None of these

Ans: Carbon dioxide.

5. The process of forming a thick oxide layer on aluminium is called

- (A) Galvanizing (C) Enrichment
(B) Anodizing (D) None of these

Ans: Anodizing

6. How many electrons are there in the outer orbit of an atom of sulphur?

- (A) 4 (B) 5
(C) 6 (D) 7

Ans : 6

7. How many chambers are present in Human heart?

- (A) 2 (B) 3
(4) 4 (D) 5

Ans: 4

8. An example of Homologous organ is

- (A) Our arm and Dog's foreleg (B) Our Teeth and elephant tusk
(C) Potato and runners of grass (D) All of the above

Ans: Our arm and Dog's foreleg

9. Which of the following are unsaturated hydrocarbons

- (A) CH_4 (B) C_2H_6
(C) C_2H_4 (D) None of these

Ans: C_2H_4

10. Which of the following compounds can be used as a fuel?

- (A) Ethanol (B) Propanol
(C) Ethanoic acid (D) none of these

Ans: Ethanol

11. The chemical formula for bleaching powder is

- (A) $\text{Ca}(\text{OH})_2$ (B) CaOCl_2
(C) CaCO_3 (D) $\text{Ca}(\text{HCO}_3)_2$

Ans: CaOCl_2

12. On decreasing the number of resistors from a parallel combination of resistors the total resistance of new combination of resistors in parallel is

- (A) Increases (B) Decreases
(C) Remains the same (D) None of these

Ans: Increases

13. Which of the following terms represent electric power in electric circuit?

- (A) I^2R (B) IR^2
(C) V^2I (D) VI^2

ans: I^2R

14. What is the commercial unit of electric energy?

- (A) Watt (B) Watt/Hour
(C) Unit (D) None of These

Ans: Watt/Hour

15. Which lens can be used for the correction of hypermetropia?

- (A) Concave Lens (B) Convex lens
(C) Sometimes concave sometimes convex (D) Cylindrical lens

Ans: Convex lens

16. Which of the following is not a receptor organ?

- (A) Ear (B) Eye
(C) Nose (D) Brain

Ans: Brain

17. The number of groups in periodic table is

- (A) 7 (B) 8
(C) 9 (D) 18

Ans: 18

18. Which phenomenon of light does the Tyndall effect show?

- (A) Reflection of light (B) Refraction of light
(C) Dispersion of Light (D) Scattering of light

Ans: Scattering of light

19. Which of the following is a malarial parasite?

- (A) Plasmodium (B) Leishmania
(C) Protozoa (D) None of these

Ans: Plasmodium

20. The pH value of pure water is

- (A) 6 (B) 7
(C) 8 (D) 9

Ans: 7

21. Which mirror is commonly used as a rear view mirror in vehicles?

- (A) Plane Mirror (B) Concave mirror
(C) Convex Mirror (D) none of these

Ans: Convex Mirror

22. According to Fleming's left hand rule, the index finger of the left hand indicates

- (A) Direction of electric force applied on a conductor (B) Direction of magnetic field
(C) Direction of flow of electric current in a conductor (D) None of these

Ans: Direction of magnetic field

23. Which of the following is an example of regeneration?

- (A) Hydra (B) Amoeba
(C) Spirogyra (D) None of these.

Ans: Hydra

24.. Which colour has the largest wavelength?

- (A) Red (B) Blue
(C) Yellow (D) Violet

Ans : Red

25. Which is not responsible for Ganga Pollution?

- (A) Fish farming in Ganga (B) washing clothes In Ganga
(C) Immersion of unburnt corpse in Ganga (D) Emission of chemical effluents in Ganga

Ans: Fish farming in Ganga

26. According to new Cartesian Sign convention, the object distance of an object placed before a mirror is taken

- (A) Positive (B) Negative
(C) Sometimes positive sometimes negative (D) None of these

Ans: Negative

27. A rectangular coil of a copper wire is rotated in a magnetic field. The direction of the induced current in the coil changes once in every –

- (A) Two rotations (B) One rotation
(C) Half rotation (D) One fourth rotation

Ans: Half rotation

28. Which of the following carbon compound is the most reactive?

- (A) CH_4 (B) C_2H_6
(C) C_2H_4 (D) C_3H_8

Ans: CH_4

29. Which of the following constitute a food chain?

- (A) Grass, Wheat and Mango (B) Grass, goat and Human
(C) Grass, Cow and Elephant (D) Grass, goat and Human

Ans: Grass, goat and Human

30. Which among these do not act as an endocrine as well as exocrine gland?

- (A) pancreas (B) Ovary
(C) Testis (D) Pituitary

Ans: pituitary

31. Which of the following is an ideal source of energy.

- (A) Coal (B) Wood
(C) Petroleum (D) Biomass

Ans: Petroleum

32. Which is a non-biodegradable waste?

- (A) Tissue paper (B) Peel of banana
(C) Thermocol (D) All of these

Ans: Thermocol

33. How many atoms of oxygen are there in one molecule of glucose?

- (A) 4 (B) 6
(C) 8 (D) 12

Ans: 6

34. How many bonds are formed between two hydrogen atoms?

- (A) Single Bond (B) Double bond
(C) Triple Bond (D) None of these

Ans: Single Bond

35. The far point for normal vision is

- (A) 25 cm (B) 25m

© 25mm

(D) Infinity

Ans: Infinity

36. Which type of mode of nutrition is found in fungi?

(A) Saprophytic

(B) Holozoic

(C) Autotrophic

(D) None of these.

Ans: Saprophytic

37. Bile juice is secreted from

(A) Pancrease

(B) Liver

(C) Small intestine

(D) None of these

Ans: Pancreases

38. Male reproductive part of flower is

(A) Stamen

(B) Pistil

(C) Petal

(D) None of these

Ans: stamen

39. Which of the following lenses have positive focal length?

(A) Concave lens

(B) Convex lens

(C) Plano- concave lens

(D) None of these

Ans: Convex lens

40. Which of the following character is inheritable?

(A) Color of eye

(B) Colour of skin

(C) Size of body

(D) Nature of Hair

Ans: Color of eye

SECTION _B

Non Objective type questions

Short answer type questions

PHYSICS

Question 1 to 8 is short answer type. Answer any four questions. Each question carries two marks.
[4×2= 8]

1. Why do we prefer convex lens as a rear view mirror in vehicles? [2]

Ans: Convex mirror are commonly used in vehicle because it gives erect image and have a wider view of as they are curved outwards.

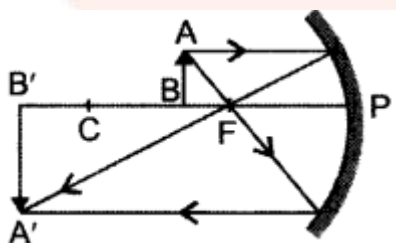
2. Why is a normal eye not see objects placed closer to 25 cm? [2]

Ans: The focal length cannot be increased or decreased beyond a certain limit due to which a healthy person cannot view clearly if the object is help less than 25cm.

3. Define refractive index. The refractive index of diamond is 2.42. What is the meaning of this statement? [2]

Ans: The extent of the change in direction that takes place in a given pair of media is Express in terms of refractive index ${}_1\mu_2$ represents the refractive index of medium 2 with respect to medium 1 when light is going from medium 1 to medium 2. The refractive index of diamond is maximum.

4. Draw a ray diagram of image of an object placed on the principal axis of a concave mirror. Write the nature, position and size of the image formed by the mirror. [2]



Object is placed beteen focus and center of curvature. The image is formed beyond the center of curvature. The image is Real, inverted and bigger in size than object.

5. What is electric circuit? Write S.I. unit of electric current.

A closed and continuous path through which electric current flows in a is known as electric circuit. It has various components including a source of current, a switch, a fuse, a battery, a load (bulb). All connected to connecting wires. The SI unit of electric current is ampere

6. An electric lamp with voltage rating 60 W; 220 v is connected to a main supply of 100 V. What current is drawn from the main supply? [2]

Ans: $P = IR$

$P = 60 \text{ Watt}$

$R = 220 \text{ V}$

$60 = I \times 220$

$I = 60/220$

$I = 0.004\Omega$

7. What is earth wire? What is its function? [2]

Ans: To avoid the risk of electric shock, the metal body of appliances is earthed. Earthing means to connect the metal case of appliances to the earth by means of a metal wire called earth wire. One end of the metal wire is buried in the earth. Appliances are connected to the earth by using the top pin of 3 pin plug. Earthing saves us from electrical shocks.

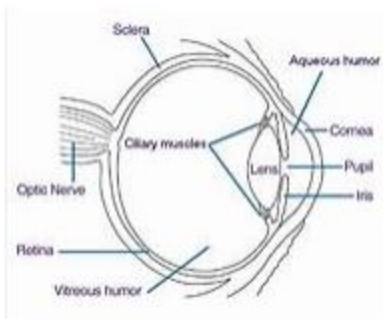
8. Why does the sun appear reddish early in the morning?

Ans: Light from the Sun near the horizon passes through a thick layer of air over a large distance in the atmosphere before reaching our eyes. Near the horizon most of the blue light and shorter wavelength are scattered away by particles; therefore, the light that reaches our eye is of longer wavelength. This gives rise to the reddish appearance of the Sun.

Question 9 and 10 are long answer type question. Answer any one of the following.

[1× 6 =6]

9. a) Draw a well labeled diagram of Human eye.



HUMAN EYE

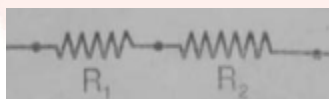
b) How does ciliary muscle accommodate focal length of the eye lens to see nearby objects and far objects?

Ans: Ciliary muscles help in accommodation by changing the curvature of eye lens. When the muscles are relax the lens becomes thin and its focal length increases. This enables us to see distant object clearly while in viewing nearby objects the lens becomes thicker and its focal length decreases.

10 a) Prove that $R = R_1 + R_2$

Where is R is the equivalent resistance of resistors R_1 and R_2 connected in series

Ans: An applied potential V produces current I through the resistor R_1 and R_2 causing a potential drop V_1 and V_2 through each resistor.



Total potential, $V = V_1 + V_2$

$$V_1 = IR_1 \quad \text{and} \quad V_2 = IR_2$$

$$V = V_1 + V_2 = IR_1 + IR_2$$

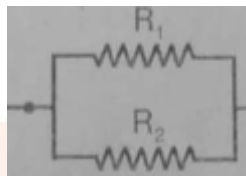
$$V = I(R_1 + R_2) \quad \text{if } R = \text{equivalent resistance and } V = IR$$

$$\text{Hence, } IR = I(R_1 + R_2)$$

$$R = R_1 + R_2$$

b) Prove that $1/R = 1/R_1 + 1/R_2$

Where R is the equivalent resistance of resistors R_1 and R_2 connected in parallel.



Ans: An applied potential V produces current I_1 in R_1 , I_2 in R_2 .

Total current, $I = I_1 + I_2$

By Ohm's law, $I_1 = V/R_1$, $I_2 = V/R_2$

If R is the equivalent resistance, then $I = V/R$

Thus, $V/R = V/R_1 + V/R_2$

Thus $V/R = V(1/R_1 + 1/R_2)$

$1/R = 1/R_1 + 1/R_2$

CHEMISTRY

Short Answer type

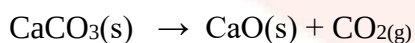
Question 11 to 17 is short answer type. Answer any four questions. Each question carries two marks.

[4 × 2 = 8]

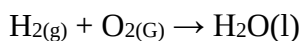
11. write one equation each for decomposition reaction and combination reaction. [2]

Ans : Decomposition reaction

Calcium carbonate on heating decomposes to give calcium oxide and carbon dioxide oxide

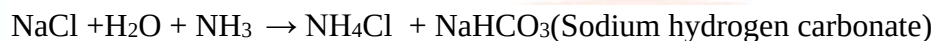


Combination reaction: Hydrogen gas and Oxygen gas react to form water

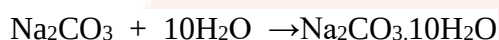


12. Differentiate between washing Soda and Baking soda . [2]

Ans: Baking soda is commonly used in kitchen for making tasty pakoda it is a mild nonproductive base chemically it is sodium hydrogen carbonate it is produced by using sodium chloride as one of the raw material .Manufacture of baking Soda is shown in the following reaction-



Washing soda is sodium carbonate .it is a white crystalline solid It's solution in water is alkaline in nature. It has the property to remove dirt and grease from the dirty clothes so it is called washing soda. It is obtained by heating baking soda.

**13. why distil water does not conduct electricity while rain water does. [2]**

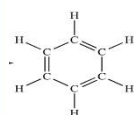
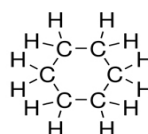
Ans: Distil water does not contain ions to conduct electricity. Rain water contains impurity so it contains ion which conducts electricity.

14. What are alloys? Give two example of alloys.

Ans: An alloy is a mixture of two or more metals or metal and non-metals that is prepared by mixing the metals in molten form and then cooling the mixture for example brass is an alloy of Copper and Tin .Iron is mixed with many metals to form alloy. Iron is very soft it is mixed with small amount of carbon and it became strong.

15. Give two ways to prevent iron from rusting

Ans: Rusting of iron is prevented by galvanizing, by making alloys, painting, greasing and tin plating

16. Draw the structure for benzene and cyclohexane.**Structure of Benzene****Structure of Cyclohexane**

17. How the electronic configuration of an atom is related to its position in Modern Periodic Table.

Ans: From the electronic configuration we can find the period and group number. The period number of an element is equal to the number of electron shells in the atom. Example if the atom has 2 electron shell then it belong to the 2nd period. The group number of an element having up to two valence electron is equal to the number of valence electron. If an element has 1 valence electron it belongs to group 1.

Long answer type questions

Question number 18 and 19 long answer type questions. Answer any one of the following. [5×1= 5].

18. What are the following reactions?

(I) Addition reaction

Ans: The reaction in which reagent completely add to a reactant without the removal of small molecules are called addition reaction. For example addition of hydrogen in the presence of catalyst like Palladium or Nickel to unsaturated hydrocarbons yields saturated hydrocarbon.

(II) Substitution reaction:

Ans: The reaction in which reagent replaces an atoms or a group of atoms from the reactants (substrate) are called substitution reaction. These are generally shown by saturated compounds in the presence of Sunlight Chlorine is added to hydrocarbon at a rapid rate. In this reaction replaces hydrogen atom one by one.

(III) Esterification reaction

(5)

Ans: When ethanol (an alcohol) reacts with acetic acid in the presence of an acid as a catalyst fruity smelling liquid called Ester is obtained. This action is called esterification.

19. Write the steps involved in the extraction of metal from their ore. [5]

Ans: Different techniques are used for extraction of metals from their ores. It depends upon the their position in the activity series which is divided into three categories

- i) Metal of low reactivity
- ii) Metal of medium reactivity.
- iii) Metal of high reactivity

Summary of these steps are given below

- i) Extraction of metal of low reactivity series-These metals being less reactive can be obtained by reducing their oxides to metal by heating alone. For example cinnabar is an ore of Mercury which when heated in air first changes into its oxide and then into Mercury metal.
- ii) Metals in the medium reactivity-The metals in the middle of activity series such as iron zinc lead copper are moderately reactive. These metals are usually present as sulphide or carbonate. In nature these sulphide or carbonate are first converted into their oxides and then metals are extracted from their oxides. Sulphides are converted into oxides by roasting and carbonates are converted into oxides by calcination. The metal oxides are reduced using suitable reducing agent.
- iii) Metal of high reactivity - Electrolytic refining is employed for these metals like sodium magnesium calcium because of their high reactivity. The salt of these metals as chlorides in molten form are electrolytes. Metals are deposited at the cathode and chlorine is liberated at the anode.

Biology

Short answer type questions.

Question number 20 to 26 short answer type question. Answer any four question. Each question carry two marks.

[4 × 2 =8]

Q.20. Differentiate between autotrophic nutrition and heterotrophic nutrition.

Ans: Some organisms use simple food material obtained from inorganic sources in the form of carbon dioxide and water. These organisms are called autotrophs and the mode of nutrition is called autotrophic nutrition. For example green plants, algae.

Organisms which utilize complex food material like carbohydrate, fat, protein to be broken down into simpler ones are called heterotrophs and the mode of nutrition is called heterotrophic nutrition.

21. Define hormone name any two plant hormones [2]

Ans: The chemical substance that are naturally produced in plants and are capable of regulating the important processes of plants are called hormones. Different plant hormones help to coordinate growth, development and response to the environment. Auxin and Gibberellins are plant hormones.

Q. 22. Give reasons for adopting contraceptive methods. [2]

Ans: By adopting contraceptive methods fertility can be controlled. Family size can be limited. It prevents the unwanted pregnancy. Sexually transmitted diseases can be controlled. Population size can be controlled.

23. Distinguish between self-pollination and cross pollination [2]

Ans: In self-pollination the Pollen from the stamen of a flower is transferred to the stigma of same flower. In cross pollination the Pollen from the stamen of a flower is transferred to the stigma of different flower.

24. What do you mean by Biological magnification? [2]

Ans: Biological magnification is a phenomenon of progressive increase in the concentration of toxicant at each successive trophic level for example pesticides like DDT sprinkled on plants enter the body of human beings through the food chain.

25. Give two suggestions to make your school environment friendly. [2]

Ans: Students hold the key to future our environment. School holds the key to educate students. There are several keys to educate students on environment friendly habits.

- 1) Plant a school garden – School garden provides fresh fruits and vegetables for meals. It helps kids their much needed outdoor time.
- 2) Make recycling bins readily accessible- The empty water bottle is a nuisance and everyone wants to get rid of it. Seeing no recycling bins in sight one should be in dilemma until a dustbin is located.

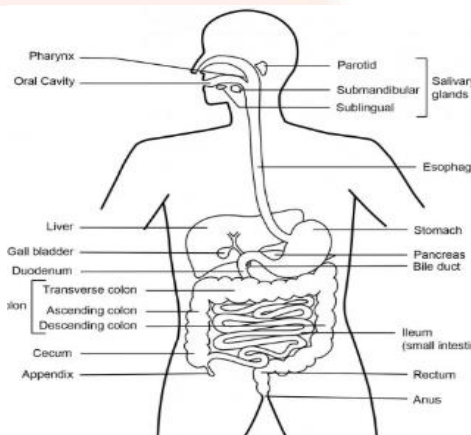
26. How is food transported in plant [2]

Ans: There is a proper system of transportation in plants. Organized conducting tubes xylem and phloem are present for transportation. Xylem and phloem transport substances from one part of the plant to another. Xylem transports water and Minerals from the soil and phloem transports product of photosynthesis from leaves to other parts of the plants.

Long answer type questions

Question number 27 and 28 are long answer type questions. Answer any one of the following.
[5 × 1 = 5]

27. Draw a well labelled diagram of human digestive system and describe the digestion of food. (5)



DIGESTIVE SYSTEM

Ans: The food we eat contains various components: carbohydrates, fats, proteins, minerals, and vitamins. Various steps that are involved in the process of digestion are following:

1. **Ingestion**- Intake of food by mouth is called ingestion. Food is mixed with saliva and masticated by teeth into smaller particles.
2. **Digestion**- The process of breaking down of complex food material into simpler form is called digestion. It is done with the help of enzymes. Starch is converted into simple sugar by salivary amylase in mouth. Proteins are converted into peptones with the help of Pepsin and renin. Bile juice emulsifies fat. Breakdown of emulsified fat is done by the action of enzyme lipase. Formation of amino acids, glucose, and fatty acids is done with the help of pancreatic and intestinal juices.
3. **Absorption**- It is the process by which food passes from the alimentary canal into the blood. All the digestive food taken up by the walls of intestine. Villi increase the surface area of large intestine.

28. Describe the structure of human brain

(5)

Ans: Brain is divided into three regions.

Forebrain, midbrain and hindbrain

Forebrain- the largest and the main thinking part of the brain which receives sensory impulse from various are ceptor is forebrain. It comprises of three main parts cerebrum, olfactory lobes and diencephalon.

Midbrain- It connect forebrain to the hindbrain .It controls movement of neck and trunk to locate sound and visual reflexes that are involved in the focusing the object .it also controls reflex moment of eye muscle Pupil size etc.

Hind Brain- It provides Connection between spinal cord and rest of the brain. It consists of three parts cerebellum, Pons and medulla oblongata.

